

SQL Code

```

1  -- =====
2  -- Description : Monthly Insurance Revenue Recapitulation Report
3  -- Purpose      : Calculate total revenue and unique buyer count per month
4  -- Source Table: sample_insurance_transaction_daily
5  -- =====
6
7  SELECT
8      -- Format transaction date (dt_id) into year-month format (YYYY-MM)
9      TO_CHAR(dt_id, 'YYYY-MM') AS transaction_month,
10
11      -- Calculate total revenue from insurance transactions for that month
12      SUM(insurance_revenue) AS total_insurance_revenue,
13
14      -- Count unique buyers (msisdn) to avoid duplicate counting
15      COUNT(DISTINCT msisdn) AS total_buyers
16
17  FROM
18      sample_insurance_transaction_daily
19
20  -- Group data by transaction month expression
21  GROUP BY
22      TO_CHAR(dt_id, 'YYYY-MM')
23
24  -- Sort report results chronologically from oldest to newest month
25  ORDER BY
26      transaction_month ASC;
27
28
29
30  -- =====
31  -- Description : Insurance Revenue Performance Report per Brand
32  -- Purpose      : Calculate total revenue and unique buyer count by brand
33  -- Sources      : sample_insurance_whitelist_daily & sample_insurance_transaction_daily
34  -- Relation     : INNER JOIN via msisdn column (Phone Number)
35  -- =====
36
37  SELECT
38      -- Brand name registered in the whitelist table
39      w.brand_name,
40
41      -- Calculate cumulative insurance revenue per brand.
42      -- COALESCE is used to replace NULL values with 0 for safe calculation.
43      SUM(COALESCE(t.insurance_revenue, 0)) AS total_insurance_revenue,
44
45      -- Count unique users/buyers (msisdn) who made transactions
46      COUNT(DISTINCT t.msisdn) AS total_buyers
47
48  FROM
49      sample_insurance_whitelist_daily AS w
50

```

```

51 -- Join whitelist data with transactions.
52 -- INNER JOIN only retrieves msisdns registered in whitelist AND have transacted.
53 INNER JOIN
54     sample_insurance_transaction_daily AS t
55     ON w.msisdn = t.msisdn
56
57 -- Group aggregation results by brand name
58 GROUP BY
59     w.brand_name
60
61 -- Sort results from highest to lowest revenue
62 ORDER BY
63     total_insurance_revenue DESC;
64
65
66
67 -- =====
68 -- Description : Customer Segmentation (User Lifecycle / RFM Analysis) - Last 90 Days
69 -- Purpose      : Classify whitelisted users into transaction behavior segments:
70 --                - Never Taker, New Buyer, Repeat Buyer, Lapsing Buyer, Lapsed Buyer
71 -- Sources      : sample_insurance_whitelist_daily &
72 --                sample_insurance_transaction_daily
73 -- =====
74 WITH whitelist_recent AS (
75     -- CTE 1: Filter users whitelisted within the last 90 days.
76     -- Uses GROUP BY msisdn to ensure 1 unique msisdn (prevents duplication
77     -- if a user has multiple brand_names in the whitelist table).
78     SELECT
79         msisdn,
80         MAX(brand_name) AS brand_name -- Take the latest/top brand_name if duplicates
81     FROM sample_insurance_whitelist_daily
82     WHERE dt_id >= CURRENT_DATE - INTERVAL '90 days'
83     GROUP BY msisdn
84 ),
85
86 txn_recent AS (
87     -- CTE 2: Aggregate last 90 days transactions per msisdn.
88     -- Calculates Recency (last transaction date) and Frequency (txn count) metrics.
89     SELECT
90         msisdn,
91         COUNT(DISTINCT transaction_id) AS txn_count_90d,
92         MAX(dt_id) AS last_txn_date
93     FROM sample_insurance_transaction_daily
94     WHERE dt_id >= CURRENT_DATE - INTERVAL '90 days'
95     GROUP BY msisdn
96 ),
97
98 merged AS (
99     -- CTE 3: Combine whitelist data with transaction statistics.
100     -- Uses LEFT JOIN so whitelisted users who never transacted are still retained.
101     SELECT
102         w.msisdn,

```

```
103     w.brand_name,
104     COALESCE(t.txn_count_90d, 0) AS txn_count_90d,
105     t.last_txn_date,
106
107     -- Calculate days difference between today and the last transaction date (Recency)
108     (CURRENT_DATE - t.last_txn_date) AS days_since_last_txn
109 FROM whitelist_recent w
110 LEFT JOIN txn_recent t
111     ON w.msisdn = t.msisdn
112 )
113
114 -- MAIN QUERY: Determine lifecycle segment based on Recency & Frequency
115 SELECT
116     msisdn,
117     brand_name,
118     txn_count_90d,
119     last_txn_date,
120     days_since_last_txn,
121
122     CASE
123         -- Segment 1: Whitelisted but never transacted in the last 90 days
124         WHEN txn_count_90d = 0 OR last_txn_date IS NULL THEN 'Never Taken'
125
126         -- Segment 2: Only 1 transaction & transaction is still recent (<= 30 days)
127         WHEN txn_count_90d = 1
128             AND days_since_last_txn <= 30 THEN 'New Buyer'
129
130         -- Segment 3: 2 or more transactions & still active (<= 30 days)
131         WHEN txn_count_90d >= 2
132             AND days_since_last_txn <= 30 THEN 'Repeat Buyer'
133
134         -- Segment 4: Transacted before, but becoming inactive (31 - 60 days since last
135         -- txn)
136         WHEN days_since_last_txn BETWEEN 31 AND 60 THEN 'Lapsing Buyer'
137
138         -- Segment 5: Transacted before, but inactive for a longer period (61 - 90 days)
139         WHEN days_since_last_txn BETWEEN 61 AND 90 THEN 'Lapsed Buyer'
140
141         ELSE 'Other'
142     END AS lifecycle_segment
143 FROM merged
144 ORDER BY
145     lifecycle_segment ASC,
146     days_since_last_txn ASC NULLS LAST;
```